Cashew Nut Growing Guide

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Kenya Agricultural & Livestock Research Organization

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Other names of cashew nuts in Kenya are:

1. Cashew nut
2. Kiswahili – Mkorosho, Mmbibo
3. Miji Kenda - Mkaju, Mmbibo, Mkorosho

1. Introduction

Origin and Distribution

Cashew nut is scientifically known as *Anacardium occidentale*, and locally known as *Mkorosho*, or *Mkanju* (Swahili). The crop originated from Northern part of South America. The Portuguese introduced cashew nut to Mozambique where it flourished forming extensive forests; eventually spread to the East African region. In Africa, this crop has spread for over 500 years either naturally or through smallholder cultivation. In Kenya, cashew nut is grown along the coastal primarily in Kwale, Kilifi, Tana River and Lamu Counties. Some production also takes place in Taita Taveta and Tharaka Nithi Counties.

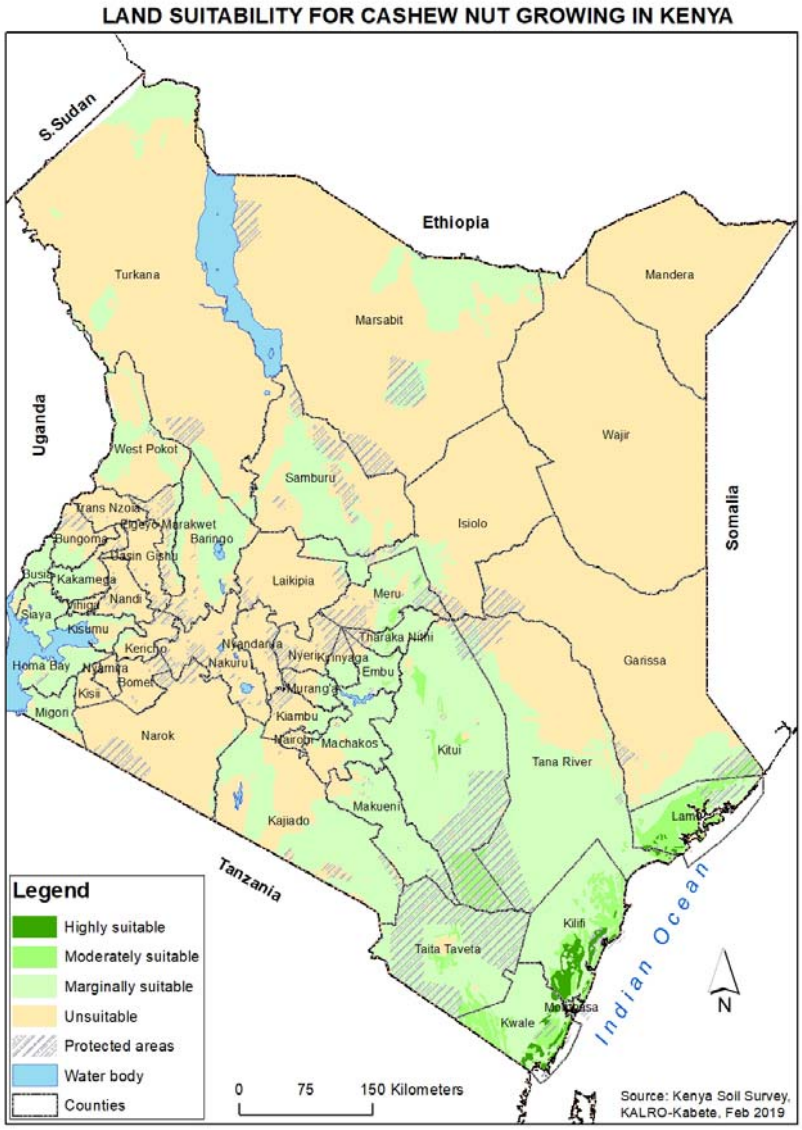

Fig. x. Cashew Nut Suitability Map

Botany

The cashew nut tree (Fig. 1) is a fast growing tree that grows to a height of 10-12m tall. Its trunk is often irregularly shaped. The leaves are spirally arranged, leathery textured, elliptic to obovate, 4 to 22cm long and 2 to 15cm broad, with a smooth margin. The flowers are produced in a panicle or corymb up to 26cm long, each flower is small, pale green at first then turning reddish, with five slender, acute petals 7 to 15mm long.

The nut is attached to the lower portion of the cashew apple which is heart like or conically shaped. The cashew nut (seed) hangs at the bottom of the apple and is c-shaped or kidney shaped. The tree has an estimated economic life span of 45 years.

Fig. 1. Mature cashew nut tree
(Source: KALRO-Mtwapa)

Fig 2. Cashew nut apple

Uses

It is grown for its fruits as well as nuts. The kernel is obtained by shelling the roasted nuts. The shells yield oil which is used as medicine, preservative and water proofing agent. The oil is also used in the manufacture of insulating varnishes and acid proof cements, tiles and inks. The raw cashew nut apple (Fig. 2) is edible and is a valuable source of sugar, minerals and vitamins. Nutritive values of cashew apple and nut are very high. Cashew nut apple is a very good source of Vitamin C. It contains 12.3% carbohydrate and 10% phosphorus and calcium. Cashew nut (kernel) contain protein (21.2%) and fat 46.9%) and carbohydrate (22.3%). It also contains comparatively good amount of minerals, iron and phosphorus. The wood and shells are used as fuel.

2. Contribution to the Kenyan Economy

Kenya produces about 15,000 metric tons valued at KES 397.4m annually against a potential of 45,000 metric tons estimated to have a value of 1 billion. Most of the cashew nut produced is from small-holdings involving about 68,000 farmers. The sub-sector has the potential to create employment through value addition and fetch the exchequer billions of shillings through exports.

3. Ecological requirements

Cashew nut does well in tropical climate with high and constant temperatures. It is frost sensitive.

Latitude

Cashew nut trees grow in a wide spectrum of climatic regions between 25˚N and 25˚S latitudes.

Altitude

It can be grown in elevations of between 0 and 1,000 metres above sea level with an ideal elevation of 600 meters above sea level. Above the altitude of 1,000 meters above sea level, cashew nut trees do not tolerate low temperatures as this interfere with reproductive cycle of the tree and leads to delayed flowering and poor yields

Temperature

Cashew nut requires hot temperatures of between 24°C and 28°C but can thrive even in temperatures of 40°C. The optimum monthly temperature for cashew nut growing is 27°C.

Rainfall

Rainfall should be between 500mm to 1200mm per year. Dry weather is required for the cashew nut plant to flower. A period of at least 4 to 5 months of dry weather after the rains is required for optimal yields. Excessive humidity leads to emergence of many fungal diseases for the cashew nut plant.

Soils

Cashew nut prefers a deep (>2m) well-drained and light textured soil. Heavy clay (including dark cracking clays), compact and hard surface setting soils hardpans or with concretions within rooting zones are not suitable for cashew nut production. Cashew nut can be described as a sand-loving plant. The optimum range of pH for this crop is between 4.5 and 6.5 and the minimum pH is 3.8. Cashew nut is not tolerant to saline soils and does not grow well in valleys, flood plains and swampy areas with poor drainage.

4. Varieties

Recommended varieties of cashew nut are A75/83 (Fig. 3), A100, A81 and A82. Cashew nut can either be propagated by seeds or grafted seedlings.

Table 1. Commercial varieties of cashew nut

Variety	Potential Yields /Kg/Tree	Kg/Acre
A81	50	1400
A82	55	1540
A75/83	70	1960
A100	60	1680

Fig. 3. New cashew nut variety introduction a 75/83 and C1 propagation (Source: KALRO-Mtwapa and Lusike Wasilwa)

5. Cashew nut propagation

Cashew nuts are usually propagated through seed or grafted seedlings (Figs. 4 and 5). In propagation by direct seeding 3 seeds are planted in the planting hole and two months after germination the weak seedlings are uprooted leaving the strong one to grow.

Propagation using grafted seedlings

Propagation through grafting starts by raising rootstock propagated from local cashew nut varieties. First step is visual selection of seeds to remove diseased or deformed seeds. The selected seeds are taken through flotation test in water. Seeds that sink are planted in pre-germination beds.

Cashew nut rootstock garden	Cashew nut seed nuts	Day 1 – soaking seed
Day 3 – place soaked seed in saw dust	Cashew nut pre–germination beds	Cashew nut seeds in pre-germination bed

Growth of pre-germinated seed	Healthy pre-germinated seedlings	Seedlings ready for grafting
Cashew nut rootstocks ready for grafting	Cashew nut scions ready for grafting	Grafted seedlings of cashew nut
Grafted seedling are individually wrapped using polyethylene tape or grafting tape or bag	Most common way to protect graft union is with grafting tape	Cashew nut seedlings ready for hardening

Fig. 4. Recommended procedure for cashew nut seedling propagation (Source: KALRO-Mtwapa and Lusike Wasilwa)

Fig. 5. Small scale propagation of cashew nut seedlings (Source: KALRO-Mtwapa)

Fig. 6. Direct potting of pre-germinated seed (Source: KALRO-Mtwapa)

The pre-germinated seeds are transplanted into potting bags when the radicals (tap root) are 2.5 –3cm after 7-10 days (Fig. 6). The recommended soil media for cashew nut seedlings is in the ratio of 2:1:150 where: 2 is two buckets of soil, 1 is one bucket of well decomposed farmyard manure and is 150gms of DAP. The seedlings are transplanted in potting bags of (6" x 9"). This composition is able to sustain the plant for averagely three months after transplanting the pre-germinated seeds in them.

Seedlings are ready for grafting when they attain more than two functional leaves and the cotyledons are still attached to the young stem. Grafting is done 3-4 weeks after transplanting into potting bags.

Scions for grafting are harvested from select mother plants. The following procedure for grafting is recommended:

a. Cut and remove the actively growing part of the rootstock, leaving two functional leaves

b. Vertically cut down the stock between the two leaves to a depth of 2.5 – 3.0cm
c. Make a wedge cut of the scion of a similar length of 2.5-3.0 cm and sharpen it
d. Insert the scion to the rootstock and tie them together
e. Cover the scion and the rootstock by wrapping with grafting tape
f. Wait for 2-3 weeks and unwrap the scion

Fig. 7. Scion preparation (Source: KALRO-Mtwapa)

Grafting tools: Tree pruner, pruning saw, secateurs- for harvesting scions, pruning grafting knife and grafting tape (Fig. 8).

Secateurs	Pruning saw	Watering can
Grafting knife	Knife sharpening stones	Grafting tape

Fig. 8. Cashew nuts propagation tools (Source: KALRO-Mtwapa and Lusike Wasilwa)

The seedlings of cashew nut are grown under shade of 60% (Fig. 9) and hardened off before planting in the orchard. It is very important not to disturb the root system during planting. Young trees should be supported with a stick for the first 2 to 3 years so that the wind will not blow the plants over.

Fig 9. 60% shade net house for cashew nut propagation (Source: KALRO-Mtwapa)

Fig. 10. Grafting cashew nut (Source: KALRO-Mtwapa)

Fig. 11. Hardening grafted cashew nut seedlings

Table 2. Location of Nurseries Propagating Cashew Nut

	County	Location	Contacts
		KALRO Centres	
1	Kilifi	KALRO-Mtwapa	P. O. Box 25 – 80109, Mtwapa
2	Kilifi	KALRO-Msabaha	P.O. Box 244-80200, Malindi
3	Kwale	KALRO-Matuga	P. O. Box 4 – 80406, Matuga
4	Lamu	KALRO-Mpeketoni	P. O. Box 15, Mpeketoni
5	Tharaka Nithi	KALRO-Marimanti	P. O. Box 27 - 60100, Embu
		ATC	
1			
2			
3			
4	Taita Taveta	Taveta	P. O. Box
		GoK Prisons	
1			

	County	Location	Contacts
2			
3			
4			
		Institutions	
1			
2			
3			
		Other Nurseries	
1	Lamu, Hindi	Masande Self Help Group	Stephen Gichobi 0714-717-928
2	Kwale & Kilifi	Ten Senses	admin@tensenses.com
3	Kilifi	Nuts & Oil crops Directorate	nocdirectorate@gmail.com
4	Tana-River	National Irrigation Board-Bura	Alfred Ngonyo: 0715361178
5			

ATC – Agricultural Training Centre
KALRO - Kenya Agricultural & Livestock Research Organization

6. Planting

Cashew nut should be planted at the beginning of the rain season. The best time for planting will depend on available soil moisture. In the Coastal lowlands, the best time for planting is between April to June.

To establish a cashew nut orchard, the following steps are recommended.

- Plough the land thoroughly, removing stems and roots
- Lay out the field by spacing holes 12m x 12m (Fig. 11)

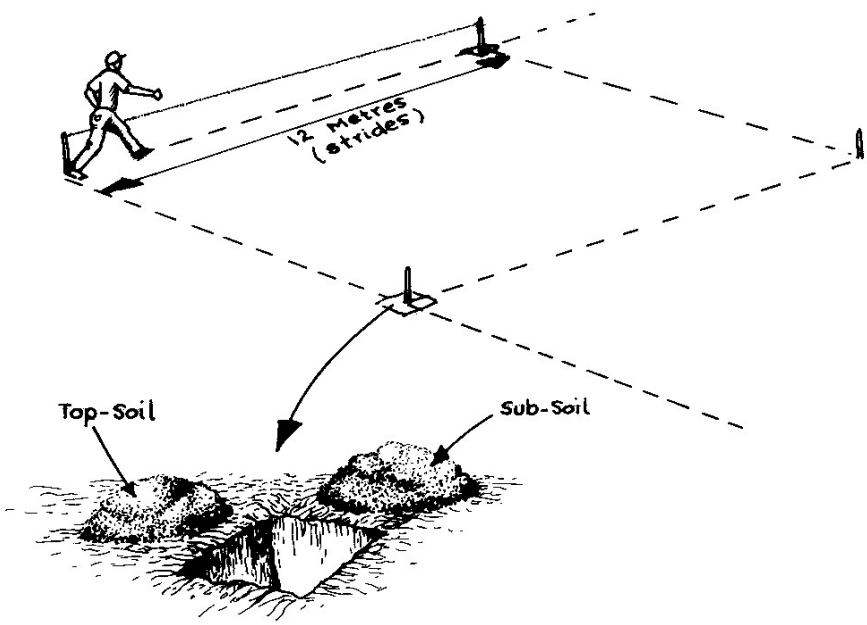

Fig. 11. Cashew nut field layout (Source: KALRO-Mtwapa)

A spacing of 6m by 6m can be adopted with intention of thinning to 12m by 12m at a later stage when the canopies start to overlap.

- Dig holes one or two months before planting. The depth should be 60cm x 60cm x 60 cm (2ft x 2ft x 2ft) (Fig. 11).
- Mix the top soil with two buckets of well decomposed farmyard manure and 150g (7.5 table spoonful) of DAP

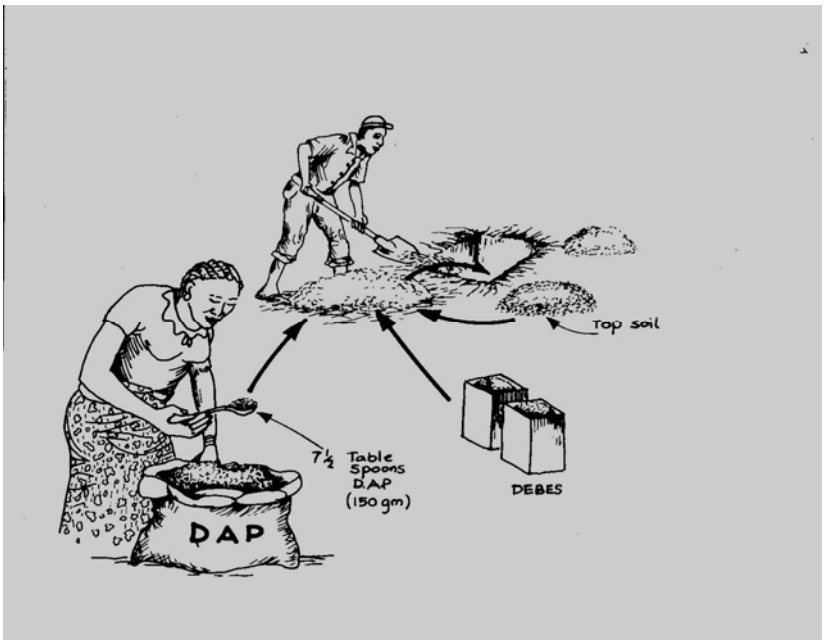

Fig. 11. Preparation of hole for planting cashew nut (Source: KALRO-Mtwapa)

- Carefully remove potting bag without disturbing the roots (Fig. 12) and plant at the centre of the hole. Plant the cashew nut seedling at the same soil level as that of the soil when the seedling was in the potting bag i.e. avoid burying the graft union. Prepare a shallow basin and cover with mulch and water after planting.

Fig. 12. Transplanting a cashew nut seedling into the field (Source: KALRO-Mtwapa)

- Water the plant every 3-4 days until it has properly taken (Fig. 13).

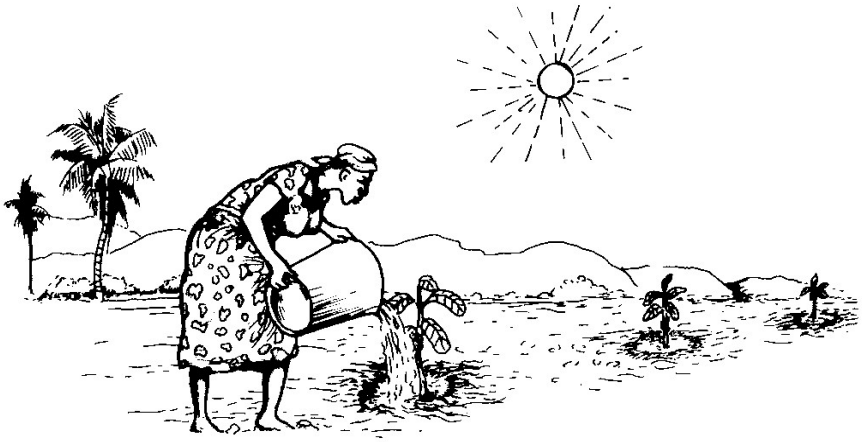

Fig. 13. Water cashew nut seedlings until the plants are established (Source: KALRO-Mtwapa)

- Intercrop with short duration perennials e.g. papaya or food crops e.g. maize, cassava, cowpeas
- Vegetables e.g. tomatoes, melons, pumpkins, chilies, egg plants (brinjals)
- Ensure that the orchard is always free from all kinds of weeds

Management of young orchard

Formative pruning

Formative pruning is the training done on the grafted plants to make the tree grow upright to a height of 1.0-1.5 m. This work should be done within the first 5 years after establishment of the cashew nut tree and is important for continuous development of the tree.

Removal of suckers

Suckers are shoots growing from the rootstock. They compete with scions for food hence they should be removed. Lateral branches on the scions suppress upright growth of the scion material. When removing horizontal shoots/branches, ensure that leaves of the main stem are not removed. Desuckering should be done continuously in year one until the tree is 1 m high.

Fire breaks

To avoid the risk of fire in the field, it is necessary to put fire breaks around the field soon after planting particularly during the dry season

Control of insect pests

Young cashew nut seedlings are susceptible to damages caused by diseases (Fig 14) and pests. It is important that the seedlings are routinely scout the seedlings and take preventative measures.

Fig. 14. Leaf spots are common on cashew nut seedlings (Source: Lusike Wasilwa)

Shading

If drought conditions occur, shade or mulch and irrigation should be provided to the seedling.

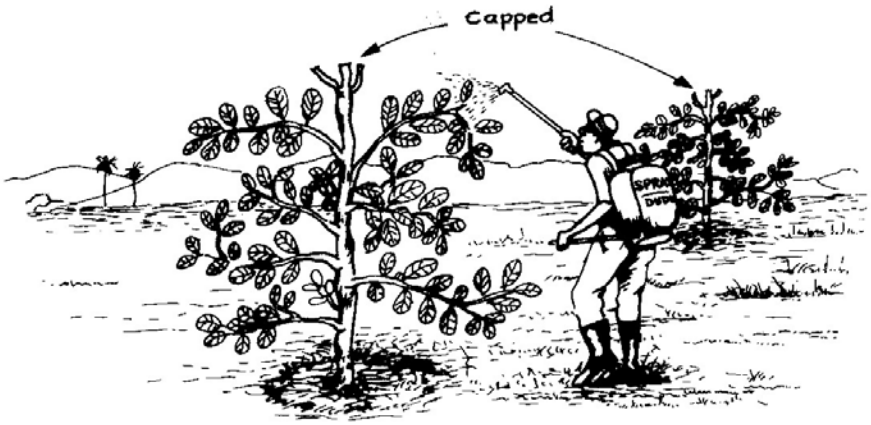

Fig. 15. Managing growth of cashew nut trees by capping (Source: KALRO-Mtwapa)

Recommended fertilizers rates

Animal manure and compost are the most desired types of fertilizers for cashew nut. However, if manure is not available or intensive production is carried out, chemical fertilizer DAP (Diammonium Phosphate, CAN (calcium ammonium nitrate and potassium sulphate can be applied. The fertilizer and manure rates recommendation are as given in the table below.

Table 3. Fertilizer and manure rates for cashew nut

Year	Manure (10-15 kg tin)		DAP (g)	CAN (g)	
	Long rains	Short rains		Long rains	Short rains
1	2	1	150	180	180
2	2	1	520	360	360
3	2	2	780	540	540
Above 3 years	2	2	780	540	540

Irrigation

Irrigation is important during establishment of young trees because it doubles the growth tempo of young trees in a dry season. Due to the deep root system the trees can survive several months without irrigation. Mature trees should receive 1,800L of water per tree every 2 weeks.

Wind control

Grass strips in the inter-rows between the tree lines are ideal to prevent erosion and should be cut regularly.

Pollination requirements

Pollination is mostly carried out by insects and after the process, it takes 6 to 8 weeks for fruits to develop. Concentration of honey bee colonies during flowering would alleviate problem of poor fruit setting.

7. Managing old cashew nut orchards

The following options are available for improving old cashew nut orchards depending on their condition.

Pruning

Pruning is done to remove unwanted plant parts. It is best done immediately after harvesting the last nuts and before flowering to give an allowance for wounds to heal.

- Remove all the dead branches
- Remove intermingling branches that prevent light penetration
- Remove branches that have been attacked by pests and diseases.
- Remove all branches that are below 1 m from the ground level
- Protect the wound with insecticidal paint
- Tools used: Pruning saws and secateurs

Selective thinning

This is removal of unwanted trees from the orchard.

- Remove trees that are too closely spaced
- Remove diseased trees
- Remove unproductive trees
- Remove undesired varieties
- Remove volunteer trees

Rehabilitation of old and unproductive trees

Pollarding

Pollarding is a pruning system in which the upper branches of a tree are removed, promoting a dense head of foliage and branches. In cashew nut severe pruning of the tree is done at a

height of about 3m to reduce canopy to allow easier spraying and harvesting. Once the tree is pollarded the canopy grows again and the tree flowers the same year.

- Remove the whole canopy at a height of 2.5 to 3.5 m from the ground
- Pollard aged but productive trees with large canopies that are difficult to manage.
- Continue with normal pruning

Coppicing

Coppicing is cutting down the tree at a height of 0.5 m to 1 m to renew the canopy

- Coppice old trees
- Unproductive trees
- Low yielding trees.
- Cut the tree at a slope to avoid water settling on the cut surface.
- Allow 3 to 4 sprouts to grow into the new canopy

Top-working

Top working is grafting of sprouts from coppiced trees. Top-work 3-4 sprouts per stem when pencil thick.

8. Pests and Disease Management

The most notable insect pests for cashew nut are the cashew nut bugs (*Helopeltis anacardii)* that suck sap from the leaves and from the young cashew nut fruit. Other insect pests include stem borers, thrips, mealybugs, weevils, caterpillars and leaf miners. Diseases include Anthracnose and powdery mildew. The table below shows the main diseases and pests found in the coastal region and recommended control measures.

Table 4. Pests and diseases of cashew nut

	Symptoms	Management
Diseases		
Powdery mildew *(Oidium anacardii)* Powdery mildew on growing tip Powdery mildew on flowers Powdery mildew on cashew nut apple	- Mildew (ash white covering) occurs on tender leaves, young shoots, flowers - Flowers are most susceptible - Seriously attacked flowers die in a few days - Can cause over 80% loss of yield - Fungus infects most at temperatures of 26-30°C and relative humidity of 90-100% - Common in cool and cloudy weather	- **Cultural control:** Physical removal of off-season sources of inoculum (particularly water shoots, off-season flowers) to delay the onset of this disease. - **Thinning:** Physical removal of trees planted close to each other so as to provide adequate space to the remaining trees. - **Pruning:** Removal of some branches/shoots within a tree to provide adequate aeration and ventilation to the tree. - **Chemical control:** Use of recommended fungicides.
Anthracnose *(Colletotricum gloesporioides)* 	- Attacks all tender parts of cashew nut - Black spots appear on attacked parts - Concentric rings on cashew nut apples	- **Cultural control:** Removal and burning of infected plant parts before the start of the cashew nut fruiting season - **Chemical control:** Use of copper-based fungicides
Die Back *Phomopsis anacardii*	- Disease causes drying up of flower stalks and young twigs of the current year - Drying up accompanied by a change of colour from green to brown, death of leaves and overhanging of flowers - Gummy exudates oozes	- **Chemical control:** Spray against cashew nut bug (*Helopeltis anacardii*) and coconut bug to prevent pre-disposing factors

	Symptoms	Management
 Dieback lesions on leaves Dieback on mature trees	out of attacked parts	
Pests		
Cashew nut bug (*Helopeltis anacardii*) 	- The bug is orange in colour - Female is larger than the male - Lays eggs on young twigs. - Population increases during rainy season - Increase in number of young twigs increases the population - The bug causes heavy damage to seedlings - Sucks sap from young leaves, shoots and young nuts and apples - A cashew nut bug attack happens suddenly and usually occurs during the rainy season	- **Cultural control:** Therefore, monitor the crop regularly to control this pest - **Biological control:** Conserve natural enemies
Cashew nut bark borer *(Mecocorynus loripes)*	-Adult is black, male is larger than female and feeds at the base of an attacked tree -Attack causes loss of water even death -Can jump to a neighbouring tree	- **Cultural control:** Cut away the bark of damaged areas of lightly infested trees and kill the larvae and pupae underneath. Repeat this once a month for six months, if required.

	Symptoms	Management
	-The larvae holes in the trunk and branches. -The female lays eggs on the dust from the holes -The pupa develops in the bark - -Heavy infestation causes the death of tree	- Destroy severely infested trees. First collect and destroy all adult weevils, then cut the tree and remove the bark to expose all larval galleries. Burn the tree to kill all larvae and pupa
Coreid bug *Pseudotheraptus wayi* 	- Adult bugs are brown in colour, densely mottled and can grow to a length of 14mm - Adult nymphs suck on flowers and developing fruits, causing them to drop or to develop cracks and sunken lesions	- **Chemical control:** Use of synthetic pyrethroids - **Biological control:** Conserve natural enemies - weaver ants nest on cashew nut trees where they deter and feed on coreid bugs
Scale insects	-	-
Mealy bugs 	- The long-tailed mealybug attacks shoots, inflorescences, apples and nuts - Affected parts appear completely white. - Trees infested during the flowering stage fail to produce fruits, whereas those infested at the nut-swelling stage produce discoloured nuts	- **Biological control:** Conserve natural enemies. Mealybugs are usually controlled by a wide range of natural enemies - **Chemical control:** Recommended insecticides can be used with white oil to control mealybugs

10. Harvesting and Postharvest Handling of Cashew Nut

Cashew nuts planted using seed begin bearing 3 to 4 years after transplanting the seedlings. The nuts should be harvested as soon as possible, especially under wet conditions and should be dried before storage.

Grafted seedlings begin bearing within 2 years of transplanting. Depending on age and maturity of plant, a tree yields between 10 to 100 kilograms of unshelled nuts per year. One hectare can thus produce between 2,000 to 5,000 kilograms of unshelled nuts per year. Although trees are production for 40 to 50 years, commercial harvesting is for 35 years.

The cashew nuts do not mature at the same time. The duration of harvest extends from 45-75 days and the nuts should be collected daily during this period. November to May is the harvesting period, with the peak harvest period from November to January. To get good quality nuts, clear the area beneath the tree, collect fallen fruits, detach the nut from the apple and dry the nuts under the sun for about 2 hours. The nuts can be graded into Fair Average Quality (FAQ) and Under Grade (UG). FAQ are well matured nuts and they should be full and well dried (12% moisture content). The colour should be grey or pale brown. They should neither be wrinkled nor spotted.

FAQ Cashew nuts

UG are well dried and mature nuts. They can be spotted but not wrinkled.

Other grading systems can be used depending on the buyer.